

A Comparative Study of Bioorthogonal Reactions with Azides

Nicholas J. Agard^{†,††}, Jeremy M. Baskin^{†,††}, Jennifer A. Prescher[†], Anderson Lo[†], and Carolyn R. Bertozzi^{†,*,§,¶,||,*}

Departments of [†]Chemistry and ^{*}Molecular and Cell Biology and [§]Howard Hughes Medical Institute, University of California, Berkeley, California 94720, and [¶]Molecular Foundry, Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720^{††}these authors contributed equally to this work.

ABSTRACT

Detection of metabolites and post-translational modifications can be achieved using the azide as a bioorthogonal chemical reporter. Once introduced into target biomolecules, either metabolically or through chemical modification, the azide can be tagged with probes using one of three highly selective reactions: the Staudinger ligation, the Cu(I)-catalyzed azide-alkyne cycloaddition, or the strain-promoted [3 + 2] cycloaddition. Here, we compared these chemistries in the context of various biological applications, including labeling of biomolecules in complex lysates and on live cell surfaces. The Cu(I)-catalyzed reaction was found to be most efficient for detecting azides in protein samples but was not compatible with live cells due to the toxicity of the reagents. Both the Staudinger ligation and the strain-promoted [3 + 2] cycloaddition using optimized cyclooctynes were effective for tagging azides on live cells. The best reagent for this application was dependent upon the specific structure of the azide. These results provide a guide for biologists in choosing a suitable ligation chemistry.

The discovery of the green fluorescent protein launched a new era in cellular biochemistry in which proteins could be monitored in complex living systems. Fluorescent protein fusions, in addition to other genetically encoded epitope tags, have been artfully employed to identify the subcellular localization, trafficking, and interaction patterns of thousands of proteins, while also facilitating their purification for molecular analysis (1). Meanwhile, biomolecules that are not directly encoded in the genome, such as glycans, lipids, and other metabolites, are not amenable to these conventional tagging technologies. Their importance as post-translational modifications and signaling molecules has placed some urgency on filling this void.

The bioorthogonal chemical reporter strategy provides a means to tag biomolecules without the requirement of direct genetic encoding (2). In this approach, a functional group (the chemical reporter) that does not interact with any biological functionality (*i.e.*, bioorthogonal) is incorporated into the target biomolecule using the cell's metabolic machinery (3–5) or delivered to a protein by virtue of its enzymatic activity (6, 7). Subsequently, the reporter is covalently tagged with an exogenous probe using a highly selective chemical reaction (8, 9). This two-step procedure has been used to modify cell surface glycans and proteins (5, 10, 11), profile protein glycosylation (12–14) and farnesylation (3), and identify proteins with a specific catalytic mechanism

using activity-based probes (6, 7). With the expansion of the bioorthogonal chemical reporter strategy into the realm of living animals (6, 7, 13, 15), the profiling and noninvasive imaging of biomarkers associated with disease progression appears imminent.

Success of the bioorthogonal chemical reporter strategy rests on the choice of the appropriate functional group and labeling reaction. The chemical reporter must be tolerated by the cellular machinery and sufficiently robust to avoid unwanted chemical or metabolic side reactions. In addition, the labeling reaction must proceed rapidly and selectively at physiological pH and temperatures. For applications using live cells or organisms, the reagents must be nontoxic.

The azide is the most versatile bioorthogonal chemical reporter. Its small size and stability in physiological settings have enabled azide-functionalized metabolic precursors to hijack the biosynthetic pathways for numerous biomolecules, including glycans (16), proteins (4, 17), lipids (3), and nucleic acid-derived cofactors (18). Three reactions have been reported for tagging azide-labeled biomolecules (Scheme 1). One of these, the Staudinger ligation (Scheme 1, reaction i), capitalizes on the selective reactivity of phosphines and azides to form an amide bond (5, 15, 19). The other two involve the reaction of azides with alkynes to give triazoles, a process that is typically very slow under ambient conditions. The Cu(I)-catalyzed azide-alkyne

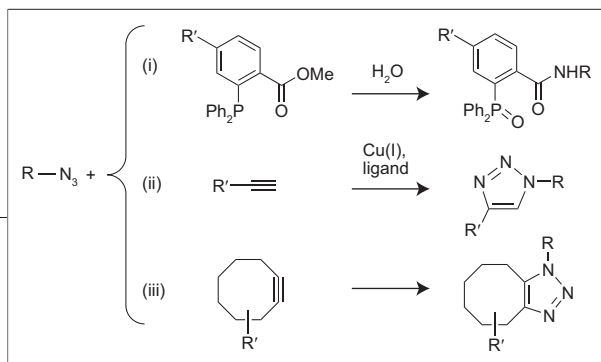
*Corresponding author,
crb@berkeley.edu

Received for review July 28, 2006
and accepted September 25, 2006.

Published online November 10, 2006

10.1021/cb6003228 CCC: \$33.50

© 2006 by American Chemical Society



Scheme 1. Bioorthogonal reactions with the azide: biomolecules containing the azide react *via* the Staudinger ligation (i), click chemistry (ii), or a strain-promoted cycloaddition (iii) to give ligated products.

cycloaddition (Scheme 1, reaction ii), also known as “click chemistry”, accelerates the reaction by use of a copper catalyst (20–22). The strain-promoted [3 + 2] cycloaddition (Scheme 1, reaction iii) removes the requirement for cytotoxic copper by employing cyclooctynes that are activated by ring strain (11, 23).

Among these options, the ideal choice of a reaction for a given biological application is not always obvious. In many cases, sensitivity is the most important parameter. Defined as the number of azides that are reacted in a given time period, sensitivity is governed by the intrinsic kinetics of the reaction and the reagent concentrations. In practice, the concentration of azides in the biological system is limited by the abundance of the target molecule and the efficiency of azide labeling. The concentration of the secondary tagging reagent (*i.e.*, the phosphine or alkyne) is typically limited by solubility and, in live cell or animal experiments, toxicity. Given these constraints, the ability to improve intrinsic reaction kinetics can be critical for optimizing a labeling strategy with respect to sensitivity. In cases where live cells or organisms are under study, biocompatibility may trump sensitivity as the most important parameter.

In order to provide a framework for choosing the optimal reaction for a given purpose, we compared the three ligations in three situations: labeling of isolated proteins, labeling of low abundance proteins from mixtures, and labeling of live cell surfaces. For high-sensitivity protein labeling, click chemistry was found to be ideal because of its superior kinetics. By contrast, live cell labeling required either the Staudinger ligation

or strain-promoted [3 + 2] cycloaddition because of their superior biocompatibility. This comparison identified a conflict between sensitivity and biocompatibility that should ultimately be resolved for an ideal labeling reaction. As a first step toward this goal, we improved the kinetics of the strain-promoted [3 + 2] cycloaddition using physical organic chemistry principles.

Initially we sought to identify the optimal reagents for each reaction with respect to intrinsic kinetics. In previous work, our attempts to improve the kinetics of the Staudinger ligation focused on increasing the electron density of the phosphine substituents (19). Although rate enhancements were observed, these were accompanied by increased rates of phosphine oxidation by air, an unwanted side reaction. Thus, the parent phosphine (Scheme 1, reaction i) remains the best reagent for biological Staudinger ligations. Other groups have directed considerable effort toward optimizing click chemistry for biological labeling reactions (7). We employed these previously reported reagents and conditions in our study.

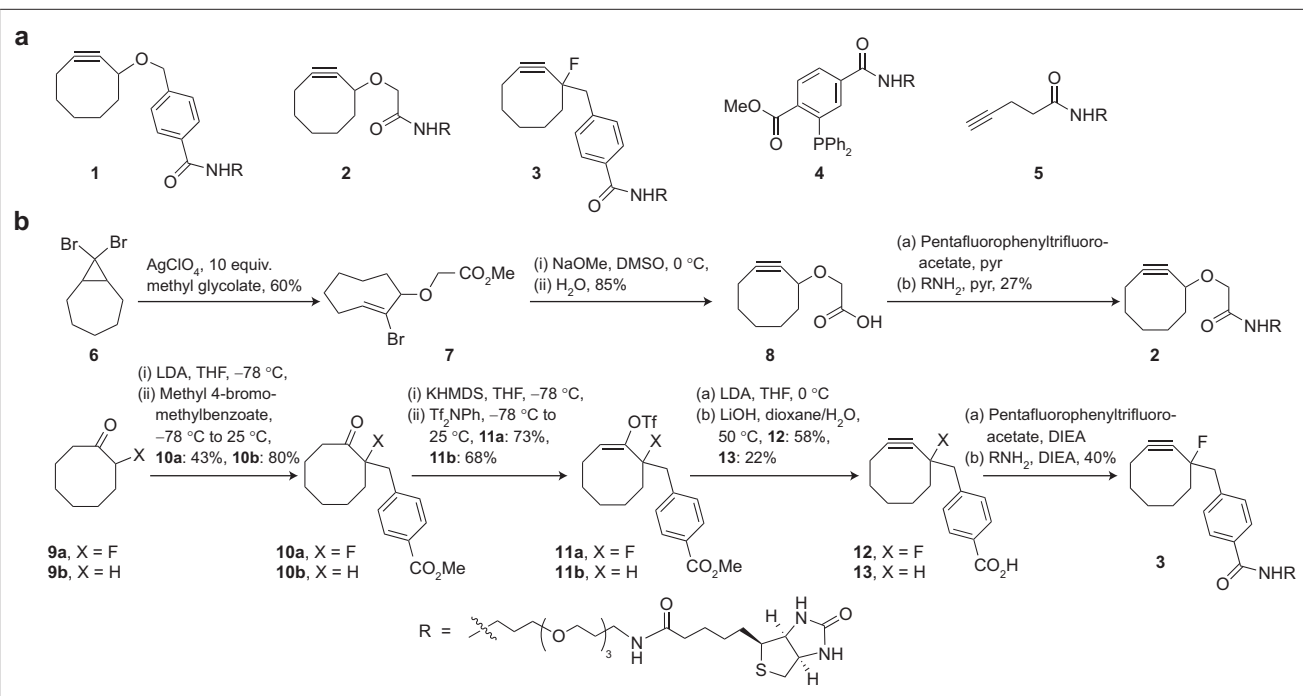
The only reaction that had not yet been explored with respect to kinetic enhancement is the strain-promoted [3 + 2] cycloaddition. Thus, we began our study by varying the substituents on the cyclooctyne scaffold. Two approaches were taken to improve upon previously reported cyclooctyne **1** (Scheme 2, panel a). First, the phenyl ring was excised (**2**) to improve the solubility of the reagent and possibly increase the rate of the reaction by decreasing steric bulk near the reactive center. In the second approach, an electron-withdrawing group (fluorine) was introduced adjacent to the alkyne (**3**) in order to lower the energy of its LUMO and promote reaction with the azide (24). As an additional benefit, this modification replaces the oxidatively labile ether link-

ages in **1** and **2** (Supplementary Scheme 1) with a stable carbon–carbon bond.

Synthesis of New Cyclooctyne Probes.

The synthesis of **2** proceeded similarly to that of compound **1** (23) (Scheme 2, panel b); silver perchlorate-mediated electrocyclic ring opening of **6** yielded a transient allylic cation that was trapped by methyl glycolate. One-pot elimination and hydrolysis of *trans*-bromocyclooctene **7** yielded the desired cyclooctyne **8**. The synthesis of cyclooctyne **3** was accomplished *via* alkylation of 2-fluorocyclooctanone (**9a**) (25) to give substituted cyclooctanone **10a**. Vinyl triflate formation and elimination, followed by saponification of the methyl ester, yielded cyclooctyne **12**. Non-fluorinated analogue **13** was synthesized *via* a similar route in order to directly assess the effect of the fluorine substituent. Biotinylation of the panel of cyclooctynes was achieved *via* formation of the pentafluorophenyl ester followed by condensation with an amine-modified biotin (26).

Kinetic Evaluation of Cyclooctynes. The relative reactivities of cyclooctynes, **8**, **12**, and **13** were determined in model reactions with benzyl azide in CD₃CN. The electron-withdrawing fluorine atom on **12** provided enhanced reactivity ($k = 4.3 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$) compared to the free acid of **1** ($k = 2.4 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$) (23) and compounds **8** ($k = 1.3 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$) and **13** ($k = 1.2 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$). In all cases, the only products observed were the triazole regioisomers as a ~1:1 mixture. Interestingly, the reaction of **12** with 2-azidoethanol in aqueous acetonitrile afforded the expected 1,5-substituted triazole, but the 1,4-isomer had undergone hydrolysis of the C–F bond to form a hydroxy-substituted product (Supplementary Scheme 2). Compound **12** in isolation was not subject to aqueous decomposition. The Staudinger ligation with benzyl azide proceeds at a similar rate in CD₃CN ($k = 2.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$) (19), while click chemistry is generally faster but subject to more complex kinetic behavior



Scheme 2. Biotinylated detection reagents used to probe for the presence of azides (a) and synthesis of new cyclooctynes (b).

because of its multiple reaction components (27). As an additional control, we investigated the chemical stabilities of cyclooctynes **8**, **12**, and **13** by incubating them in aqueous acid, aqueous base, β -mercaptoethanol, or phosphate-buffered saline (PBS) (see Supporting Information for details). In all cases, no decomposition was observed by ^1H NMR and ^{19}F NMR, where applicable.

Protein Labeling. We next sought to compare the reactions in the context of biological labeling experiments. The least demanding situation involves modification of a purified azide-labeled biomolecule. Thus, we expressed dihydrofolate reductase (DHFR) in which methionine residues were replaced with the unnatural amino acid azidohomoalanine (DHFR-N₃) (4, 28). First, the time dependencies of the reactions of compounds **1–5** with DHFR-N₃ were compared. DHFR-N₃ was incubated with **1–5** (100 μM) for 4, 12, or 24 h, and reaction progress was monitored by Western blot using anti-biotin antibody-HRP conjugate (Figure 1, panel a). For click chemistry, the relative concentrations of **5**, triazoly ligand, tris-carboxyethylphosphine (TCEP), and CuSO₄ were held at a ratio of 1:1:10:10, as optimized by Cravatt *et al.* (7). Preliminary investigations found that urea and ionic detergents used to solubilize the protein inhibited click chemistry and the Staudinger

ligation, respectively (data not shown). Consequently, Staudinger ligations were performed in 8 M urea while the remainder of the ligations were run in 1% SDS. All of the protein labeling reactions were highly specific at 37 °C, as native DHFR showed no detectable biotinylation. However, nonspecific biotinylation was observed for both the strain-promoted cycloadditions and click chemistry when the reactions were boiled (data not shown). All five reagents tested showed time-dependent protein labeling that was consistent with their kinetics in model reactions (Figure 1, panel a). Click chemistry afforded the highest sensitivity of labeling, with maximal labeling observed by 4 h.

Next, the concentration dependence of the reactions was compared (Figure 1, panel a). The reactions were performed with 50, 100, or 200 μM reagent for a period of 8 h. All three ligations exhibited labeling proportional to reagent concentration. Interestingly, the large shift in apparent molecular weight of DHFR-N₃ in the click chemistry lanes underscores the higher-order concentration dependence that this ligation obeys because of its multicomponent nature (27). This observation is particularly relevant for applications that require low concentrations of labeling reagent to avoid toxicity. In addition, in living animals, where the site of injection can be far removed from the tar-

geted biomolecule, trafficking can limit the concentration of reagents near the azide.

A more stringent test of the reactions' bioorthogonality is their ability to specifically label azides in the presence of complex mixtures, a feature essential for enrichment from lysates in preparation for proteomic analysis. To simulate a labeled lysate, we combined DHFR-N₃ with 200 μg of *Escherichia coli* lysate (total protein concentration of 10 mg/mL). These mixtures were then labeled with reagents **1–5** (200 μM , 8 h). Specific protein labeling was observed with all five reagents, with click chemistry affording the highest sensitivity, consistent with the trend observed using purified protein.

Live Cell Labeling. Having demonstrated selective labeling of proteins within complex lysates, we sought to extend the comparison to labeling of living cells. Jurkat cells were grown in the presence of 25 μM peracetylated *N*-azidoacetylmannosamine (Ac₄ManNAz) for 2–3 d, leading to the metabolic incorporation of the corresponding *N*-azidoacetyl sialic acid (SiaNAz) into their cell surface glycoproteins (5, 29). The cells were then treated with compounds **1–5** (50 or 100 μM for 1 h), incubated with FITC-avidin, and analyzed by flow cytometry (Figure 1, panel c). We observed significant cell death in the presence of the click reagents (Supplementary Figure 1), which

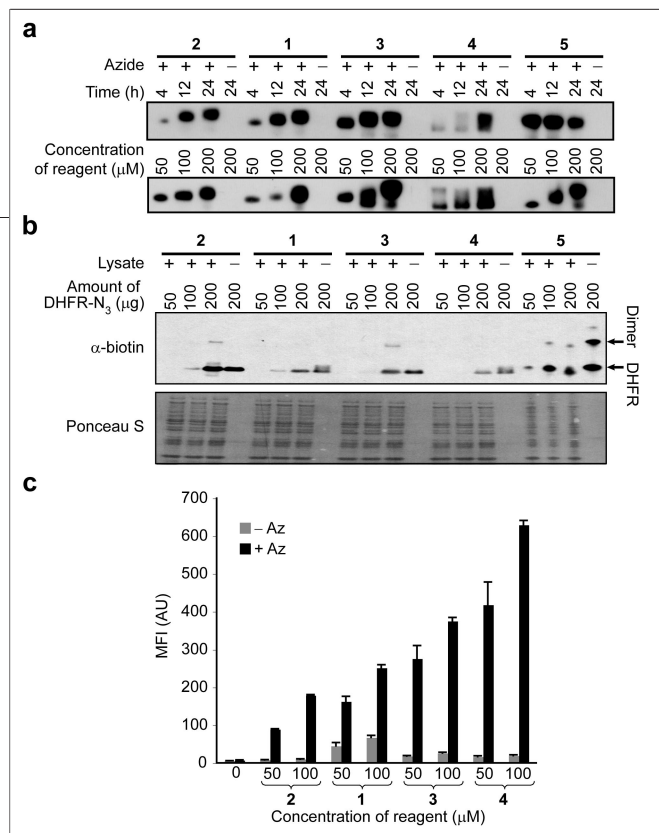


Figure 1. Relative labeling efficiencies of the reactions on isolated proteins, cell lysate, and live cells. **a)** DHFR (200 μ g) grown in the presence (+) or absence (-) of azidohomoalanine was labeled with biotinylated reagents 1–5 under given conditions followed by Western blot. Top: Time course analysis at 100 μ M reagent. Bottom: Concentration dependence over 8 h. The protein bands broaden and increase in molecular weight over time and concentration, reflecting multiple sites of modification. **b)** DHFR-N₃ in the presence (+) or absence (-) of 10 mg mL⁻¹ *E. coli* cell lysate is labeled with 200 μ M reagent for 8 h. Top: Labeled proteins are detected by Western blot analysis. Labeling of DHFR-N₃ and its dimer was found to be largely lysate-independent for each of these reactions. Bottom: Total protein content was determined using Ponceau S. **c)** Jurkat cells grown for 3 d in the presence (+ Az) or absence (- Az) of peracetylated *N*-azidoacetylmannosamine (25 μ M) were labeled for 1 h with 0, 50, or 100 μ M reagent followed by secondary labeling with FITC-avidin. The resulting mean fluorescence intensity (MFI) of the cell populations was determined by flow cytometry. Error bars represent the standard deviation from three separate experiments.

we attribute to the high concentration of CuSO₄, as previously suggested in the context of an *E. coli* labeling study (11, 20). Cyclooctyne reagents **1–3** labeled cells in proportion to their rate constants in the model reaction, while phosphine **4** showed a higher degree of labeling than expected based on its relative rate of reaction with benzyl azide. To determine the basis of this discrepancy, we performed additional model reactions of phosphine **4** and cyclooctyne **1** with α -azido acetamides that better mimic the reactivity of SiaNAz. In these reactions, the phosphine outperformed the cyclooctyne by ~2-fold (Supplementary Table 1). These results suggest that the Staudinger ligation is more efficient for labeling azides bearing electron-withdrawing or resonance stabilizing groups. The strain-promoted cycloaddition is relatively insensitive to the electronics of the azide and is more efficient than the Staudinger ligation with unactivated alkyl azides.

Reaction Guide for Specific Applications.

Collectively, these experiments provide a guide for choosing the optimal ligation chemistry for specific applications (Table 1). Each of the reactions is competent to label isolated biomolecules. Click chemistry is the

most efficient application, although some groups have reported difficulty in separating intact modified biomolecules from catalytic copper (22). The residual heavy metal was found to interfere with analysis by mass spectrometry and could alter the activity of the modified protein. For situations where copper is a concern and purification by chromatography is prohibitive, either phosphine **4** or cyclooctyne **3** is an appropriate choice.

For proteomic applications, click chemistry is the clear choice. Its superior sensitivity should provide the most efficient detection of low abundance species. Furthermore, the difficulties encountered when intact proteins bind copper can be eliminated by trypsinization and LC (6). The only caveat to this approach is that the chemistry is not compatible with all detergents. For rare cases where labeling in the presence of specific detergents is necessary, cyclooctyne **3**

most efficient reaction for this

is the best alternative, as phosphine **4** has also shown detergent sensitivity.

For labeling of live cells, the Staudinger ligation and strain-promoted [3 + 2] cycloadditions are both suitable choices, with the best reaction being dictated by the nature of the azide. At this point, the toxicity of the copper catalyst necessary for click chemistry limits its utility for labeling live cells. The Staudinger ligation has already been shown to perform in living mice without discernible toxicity (13, 15). An interesting future direction is to explore whether the cyclooctynes are amenable to such applications.

An important conclusion from this study is that a need remains for azide ligation reactions that are both highly sensitive and biocompatible. Within the current repertoire of reactions, these attributes are in conflict. Optimization of the cycloaddition reagents might be possible *via* synthesis of tight-binding copper ligands to mitigate click

TABLE 1. Selection of reaction depends on application^a

Reaction	Optimal for labeling
Staudinger ligation	Surfaces of live cells; live organisms
Click chemistry	Proteomic samples
Strain-promoted [3 + 2] cycloaddition	Surfaces of live cells

^aReagents used in each reaction are shown in Scheme 1.

chemistry's cytotoxicity or further substitutions on the cyclooctyne scaffold to increase its rate. Although attempts to increase the rate of the Staudinger ligation have led to increases in non-specific oxidation, its increased selectivity at elevated temperatures might enable rate enhancement by localized heating (*i.e.*, via focused ultrasound) (30).

METHODS

New Compounds. Synthetic methods and characterization of new compounds is provided in Supplementary Methods.

Kinetic Evaluation of Cyclooctynes. The rates of reaction between cyclooctynes and various azides in CD₃CN were monitored by the disappearance of starting materials and appearance of the two regioisomeric products in the ¹H NMR spectrum. Second-order rate constants for the reaction were determined by plotting the 1/[reagent] vs time, followed by subsequent analysis by linear regression. The rate constants correspond to the determined slope. For further details, please see the Supporting Information.

Western Blot Analysis of DHFR-N₃. For time-dependent Western blot analysis of DHFR and DHFR-N₃, 200 ng portions of protein samples were incubated in 20 μ L of **1–4** (100 μ M final concentration) or **5** (100 μ M final concentration) with 1 mM TCEP, 100 μ M tris-triazolyl ligand (TBTA) and 1 mM CuSO₄ for the indicated periods of time. Prior to electrophoresis, samples were incubated with an equal volume of 100 mM 2-azidoethanol in 2X SDS-PAGE loading buffer for 8 h at RT (to quench unreacted **1–5**). Samples were subject to SDS-PAGE analyzed by Sypro Ruby Red or transferred to nitrocellulose membranes and detected by Western blot analysis with anti-biotin-HRP antibody. Concentration-dependent labeling was determined in the same manner, but with the indicated concentration of **1–4** or **5** with triazolyl ligand TBTA, TCEP and CuSO₄ in a 1:1:10:10 ratio for 8 h.

For labeling in the presence of complex mixtures, the indicated amount of protein was incubated with 200 μ g (10 mg mL⁻¹ of soluble protein from *E. coli* lysate and buffers as above (final concentration of **1–5** was 200 μ M). The extent and specificity of labeling were determined by Western blot analysis as described above. Detailed experimental procedures for Western blot experiments are included in the Supporting Information.

Cell Surface Azide Labeling and Detection. Jurkat cells bearing azides were treated with 0–100 μ M of the biotinylated probes **1–4** in labeling buffer (PBS, pH 7.4 containing 1% (v/v) fetal calf serum) and analyzed by flow cytometry as previously described (30). For all flow cytometry experiments, data points were collected in triplicate and are representative of three separate experiments. For further details, see the Supporting Information.

Acknowledgment: This research was supported by a grant from the National Institutes of Health (GM58867). The authors thank Isaac Carrico and Jason Rush for materials and the Bertozzi lab for helpful discussions. J.A.P. was supported by a Howard Hughes Medical Institute predoctoral fellowship and J.M.B. was supported by a National Defense Science and Engineering Graduate fellowship.

Supporting Information Available: This material is available free of charge via the Internet.

REFERENCES

- Zacharias, D. A., and Tsien, R. Y. (2006) Molecular biology and mutation of green fluorescent protein, *Methods Biochem. Anal.* 47, 83–120.
- Prescher, J. A., and Bertozzi, C. R. (2005) Chemistry in living systems, *Nat. Chem. Biol.* 1, 13–21.
- Kho, Y., Kim, S. C., Jiang, C., Barma, D., Kwon, S. W., Cheng, J., Jaunbergs, J., Weinbaum, C., Tamanoi, F., Falck, J., and Zhao, Y. (2004) A tagging-via-substrate technology for detection and proteomics of famesylated proteins, *Proc. Natl. Acad. Sci. U.S.A.* 101, 12479–12484.
- Klick, K. L., Saxon, E., Tirrell, D. A., and Bertozzi, C. R. (2002) Incorporation of azides into recombinant proteins for chemoselective modification by the Staudinger ligation, *Proc. Natl. Acad. Sci. U.S.A.* 99, 19–24.
- Saxon, E., and Bertozzi, C. R. (2000) Cell surface engineering by a modified Staudinger reaction, *Science* 287, 2007–2010.
- Speers, A. E., Adam, G. C., and Cravatt, B. F. (2003) Activity-based protein profiling in vivo using a copper(I)-catalyzed azide-alkyne [3 + 2] cycloaddition, *J. Am. Chem. Soc.* 125, 4686–4687.
- Speers, A. E., and Cravatt, B. F. (2004) Profiling enzyme activities in vivo using click chemistry methods, *Chem. Biol.* 11, 535–546.
- Hang, H. C., and Bertozzi, C. R. (2001) Chemoselective approaches to glycoprotein assembly, *Acc. Chem. Res.* 34, 727–736.
- Link, A. J., Mock, M. L., and Tirrell, D. A. (2003) Non-canonical amino acids in protein engineering, *Curr. Opin. Biotechnol.* 14, 603–609.
- Mahal, L. K., Yarema, K. J., and Bertozzi, C. R. (1997) Engineering chemical reactivity on cell surfaces through oligosaccharide biosynthesis, *Science* 276, 1125–1128.
- Link, A. J., Vink, M. K. S., Agard, N. J., Prescher, J. A., Bertozzi, C. R., and Tirrell, D. A. (2006) Discovery of aminoacyl-tRNA synthetase activity through cell-surface display of noncanonical amino acids, *Proc. Natl. Acad. Sci. U.S.A.* 103, 10180–10185.
- Vocadlo, D. J., and Bertozzi, C. R. (2004) A strategy for functional proteomic analysis of glycosidase activity from cell lysates, *Angew. Chem., Int. Ed.* 43, 5338–5342.
- Dube, D. H., Prescher, J. A., Quang, C. N., and Bertozzi, C. R. (2006) Probing mucin-type O-linked glycosylation in living animals, *Proc. Natl. Acad. Sci. U.S.A.* 103, 4819–4824.
- Sprung, R., Nandi, A., Chen, Y., Kim, S. C., Barma, D., Falck, J. R., and Zhao, Y. (2005) Tagging-via-substrate strategy for probing O-GlcNAc modified proteins, *J. Proteome Res.* 4, 950–957.
- Prescher, J. A., Dube, D. H., and Bertozzi, C. R. (2004) Chemical remodelling of cell surfaces in living animals, *Nature* 430, 873–877.
- Dube, D. H., and Bertozzi, C. R. (2003) Metabolic oligosaccharide engineering as a tool for glycobiology, *Curr. Opin. Chem. Biol.* 7, 616–625.
- Chin, J. W., Cropp, T. A., Anderson, J. C., Mukherji, M., Zhang, Z., and Schultz, P. G. (2003) An expanded eukaryotic genetic code, *Science* 301, 964–967.
- Weller, R. L., and Rajsiki, S. R. (2005) DNA methyltransferase-mediated click chemistry, *Org. Lett.* 7, 2141–2144.
- Lin, F. L., Hoyt, H. M., Van Halbeek, H., Bergman, R. G., and Bertozzi, C. R. (2005) Mechanistic investigation of the Staudinger ligation, *J. Am. Chem. Soc.* 127, 2686–2695.
- Link, A. J., and Tirrell, D. A. (2003) Cell surface labeling of *Escherichia coli* via copper(I)-catalyzed [3 + 2] cycloaddition, *J. Am. Chem. Soc.* 125, 11164–11165.
- Link, A. J., Vink, M. K. S., and Tirrell, D. A. (2004) Presentation and detection of azide functionality in bacterial cell surface proteins, *J. Am. Chem. Soc.* 126, 10598–10602.
- Wang, Q., Chan, T. R., Hilgraf, R., Fokin, V. V., Sharpless, K. B., and Finn, M. G. (2003) Bioconjugation by copper(I)-catalyzed azide-alkyne [3 + 2] cycloaddition, *J. Am. Chem. Soc.* 125, 3192–3193.
- Agard, N. J., Prescher, J. A., and Bertozzi, C. R. (2004) A strain-promoted [3 + 2] azide-alkyne cycloaddition for covalent modification of biomolecules in living systems, *J. Am. Chem. Soc.* 126, 15046–15047.
- Nguyen, L. T., De Proft, F., Dao, V. L., Nguyen, M. T., and Geerlings, P. (2003) A theoretical approach to the regioselectivity in 1,3-dipolar cycloadditions of diazoalkanes, hydrazoic acid and nitrous oxide to acetylenes, phosphalkynes and cyanides, *J. Phys. Org. Chem.* 16, 615–625.
- Nyffeler, P. T., Duron, S. G., Burkhart, M. D., Vincent, S. P., and Wong, C.-H. (2005) Selectfluor: mechanistic insight and applications, *Angew. Chem., Int. Ed.* 44, 192–212.
- Wilbur, D. S., Hamlin, D. K., Vessella, R. L., Stray, J. E., Buhler, K. R., Stayton, P. S., Klumb, L. A., Pathare, P. M., and Weerawarna, S. A. (1996) Antibody fragments in tumor pretargeting. Evaluation of biotinylated Fab' colocalization with recombinant streptavidin and avidin, *Bioconjugate Chem.* 7, 689–702.
- Rodionov, V. O., Fokin, V. V., and Finn, M. G. (2005) Mechanism of the ligand-free CuI-catalyzed azide-alkyne cycloaddition reaction, *Angew. Chem., Int. Ed.* 44, 2210–2215.
- Mangold, J. B., Mischke, M. R., and LaVelle, J. M. (1989) Azidoalanine mutagenicity in Salmonella: Effect of homology and alpha-methyl substitution, *Mutat. Res.* 216, 27–33.
- Saxon, E., Luchansky, S. J., Hang, H. C., Yu, C., Lee, S. C., and Bertozzi, C. R. (2002) Investigating cellular metabolism of synthetic azidosugars with the Staudinger ligation, *J. Am. Chem. Soc.* 124, 14893–14902.
- Wu, F. (2006) Extracorporeal high intensity focused ultrasound in the treatment of patients with solid malignancy, *Minim. Invasive Ther. Allied Technol.* 15, 26–35.